DIYA SINGH

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EDUCATION

M.S. Computer Science, Purdue University **Aug 2025 - May 2027**
Coursework: Artificial Intelligence, Data Mining, Statistical ML, Operating Systems, Parallel Computing, Human-Computer Interaction

B.S. Computer Science, Purdue University **Aug 2022 - May 2026**
Minor: Mathematics, Art & Design | Concentrations: Machine Intelligence, Systems Software, Algorithmic Foundations, Computer Graphics
Coursework: Data Engineering, Advanced Algorithms, Operating Systems, Compilers, Systems Programming, Computer Networks, Security

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, Bash scripting, JavaScript, TypeScript
Frameworks: TensorFlow, PyTorch, scikit-learn, pandas, NumPy, Langchain, Hugging Face, FastAPI, SQLAlchemy, React
Systems / Infra: Linux/Unix, TCP/IP, Docker, Terraform, GitHub Actions, Git, Postgres, MySQL, Google Cloud Platform, Azure
Certifications: ML Specialization (Stanford/DeepLearning.AI); Google AI Professional Certificate; Build Infrastructure with Terraform on Google Cloud; ML Summer School 2026 (Cohere Labs); AWS AI & ML Scholar (Future AI Scientist Nanodegree)

EXPERIENCE

AI & ML Automation Intern, Envita Solutions **May 2026 - Aug 2026**
- Built production ML/LLM pipeline classifying, extracting structured service requests from unstructured emails across 40+ mailboxes.
- Containerized deployed pipeline on Azure using Python and Docker, scaling throughput to 2,000+ emails per minute.
- Authored 180+ automated unit and integration tests and served as on-call engineer, triaging failures to minimize downtime.
- Improved classification accuracy 60% to 99.12% and extraction accuracy to 90% via human-in-the-loop review and structured logging.

Product Development Intern, FractalHive **Jul 2025 - Sep 2025**
- Built responsive audit web app (Python, React, Supabase) for an education-sector client, cutting manual processing time 30%.
- Designed a validated, access-controlled database schema and role-specific dashboards for a B2B banking platform.
- Provided on-call support, diagnosing and resolving production bugs across manager and underwriter workflows.

Discovery Undergraduate Research Intern (DUIRI), Purdue University **Jan 2025 – Present**
- Engineered an AI-powered Walkability Assistant app informed by analyzing 2M+ user reviews with LDA topic modeling in Python; led user-centered design and prototype testing, integrating feedback for a 25% increase in user engagement.

Machine Learning Intern, Haleon - Centrum Multivitamins **Jul 2024 - Aug 2024**
- Owned full-stack design and implementation (Flutter + Python + Firebase) for a nutrient deficiency diagnostic app end to end, iterating weekly with stakeholders in agile sprints.

Machine Learning Intern, nRoad **May 2024 - Jul 2024**
- Built finance document-parsing model and REST API (FastAPI, SQLAlchemy, MySQL) in a microservice architecture.

Undergraduate Researcher, Purdue University - Dr. Wenzhuo Wu's Lab **Aug 2023 - Jan 2024**
- Developed tinyML models for wearable sensors, optimizing performance under tight hardware resource constraints.
- Built Unity-based digital screen printer model, cutting resource waste 30% and testing time 35%; presented at Purdue research expo.

Head Undergraduate Teaching Assistant, CS180, Purdue University **Aug 2025 - May 2026**
- Led weekly lecture/lab sessions for 400+ students in CS180 (Problem Solving & OOP); coordinated the grading team and mentored students in debugging, code optimization, and software engineering best practices.

Undergraduate Teaching Assistant, CS381 & CS182, Purdue University **Aug 2023 - May 2026**
- TA for Analysis of Algorithms and Discrete Mathematics, leading office hours covering asymptotic analysis, dynamic programming, and graph algorithms.

SELECTED PROJECTS

CharakBloom: AI-Powered PCOS Management App **May 2026**
- Built iOS app (React Native, Gemini 2.5 Flash) generating personalized meal, workout plans by cycle phase, pantry, and equipment.
- Implemented exponential decay-weighted averaging cycle prediction across prior cycles, with confidence intervals on each forecast.
- Architected a single daily AI call with static fallback, keeping the app usable offline and within free-tier API limits.

Live Streaming Application **Apr 2025 - May 2025**
- Engineered a real-time live-streaming system using Docker containers, hosting a public server on a Google Cloud Platform VM.
- Automated performance analytics and efficient pcap data processing in Python under different network conditions.

Animal Detection & Segmentation Under Corruptions **Dec 2025**

- Fine-tuned and benchmarked YOLO, Mask R-CNN, and RT-DETR under systematic Gaussian blur and JPEG compression at three severity levels, rather than evaluating only on clean data.
- RT-DETR achieved the highest baseline mAP@0.5 (0.788) and the lowest degradation under high-blur conditions, informing which model was most robust to real-world input corruption.

PUBLICATIONS

- "What walking-App Reviews Make Visible: A Multidimensional Analysis of Digitally Mediated Walking" - Manuscript in Progress

LEADERSHIP & HONORS

- Design Team Lead, BoilerMake Hackathon Executive Board (Feb 2023 - May 2026)
- Graphic Design Lead, Boilerexams (Aug 2023 – May 2026)
- Women in Science Program – Jandos Scholar, Mentor, & Social Media Committee (Aug 2022- May 2026)
- Director of Technology, Purdue Student Government (2025)
- Ford Motors Company Scholarship (2024); Purdue CS Corporate Partners Scholarship (2025); Pillars of Excellency Scholarship (2024)
- Memberships: Rewriting The Code, CodePath, Society of Women Engineers – Attending SWE26